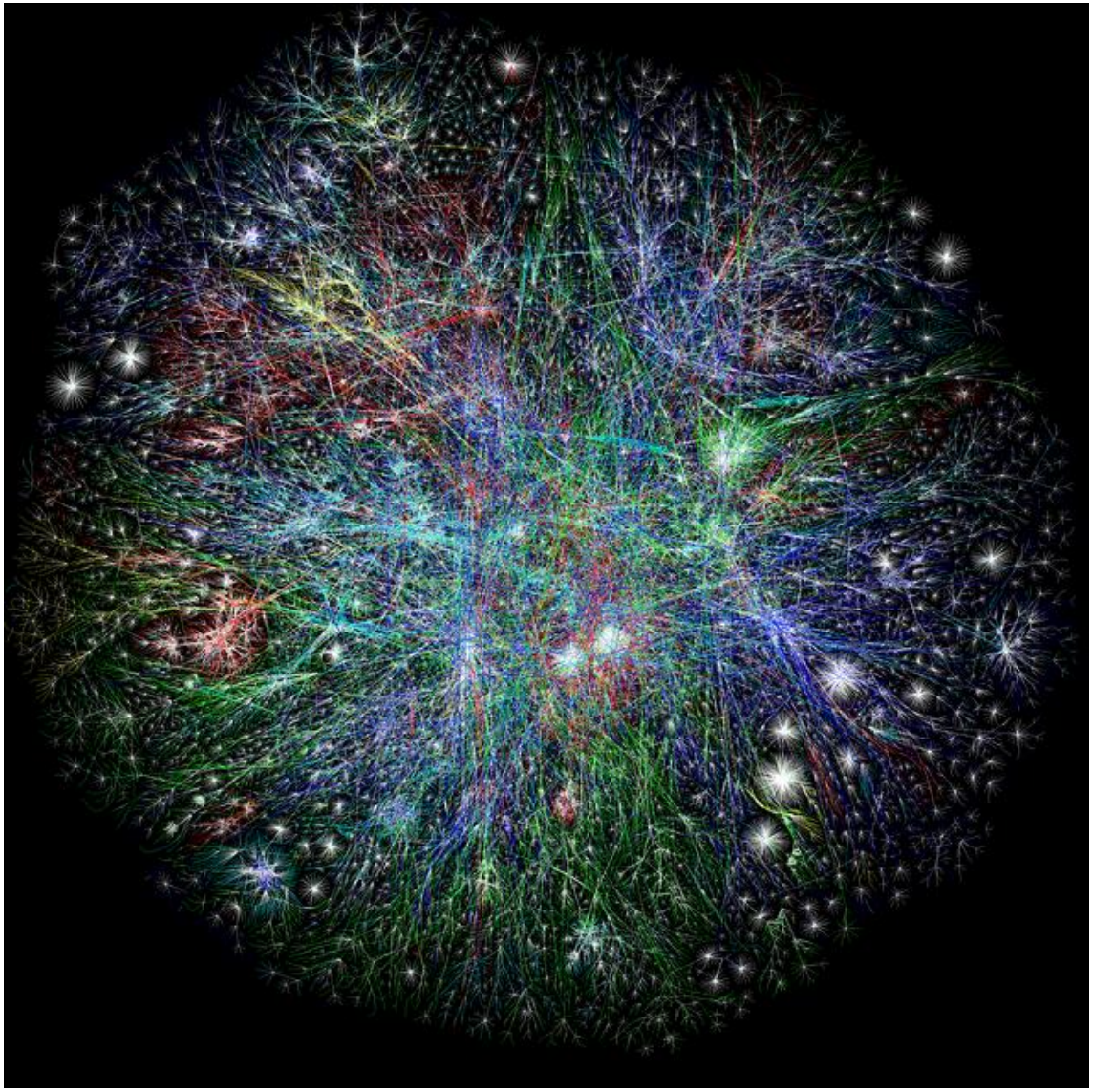
COMMUNITY DETECTION

Graph Data!

Internet



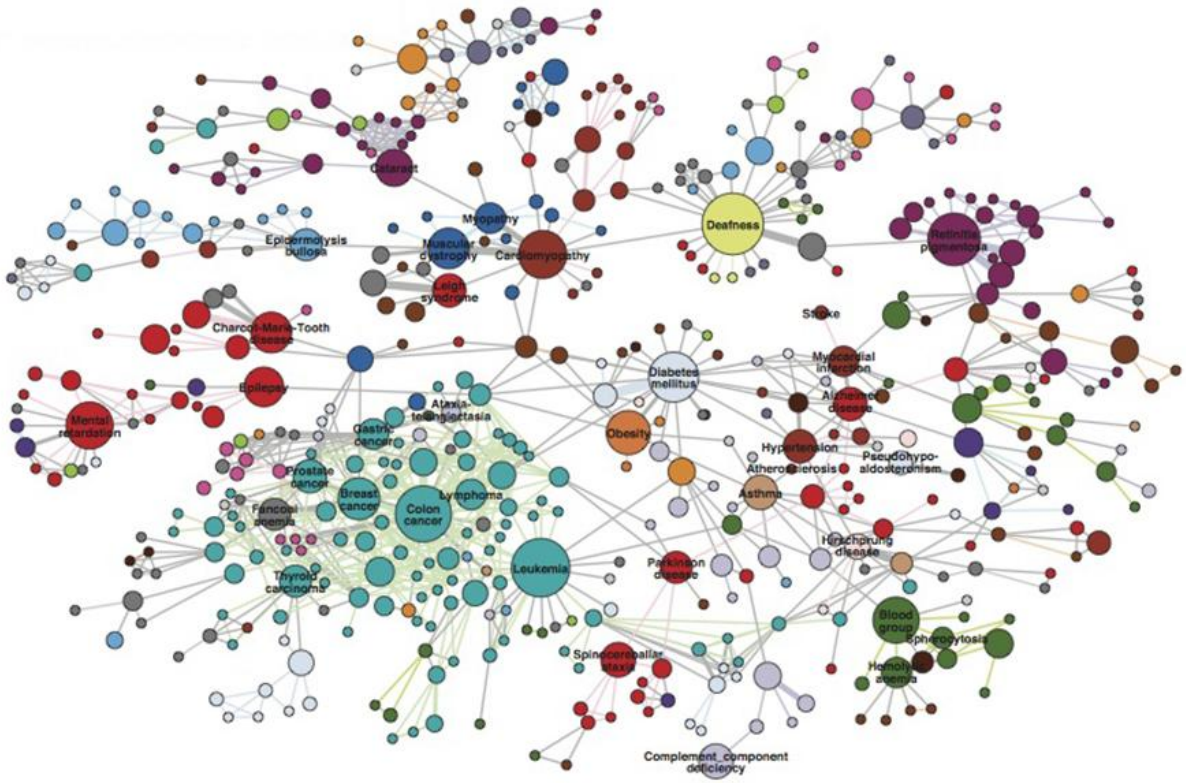
Graph Data: Social Networks

Facebook social graph

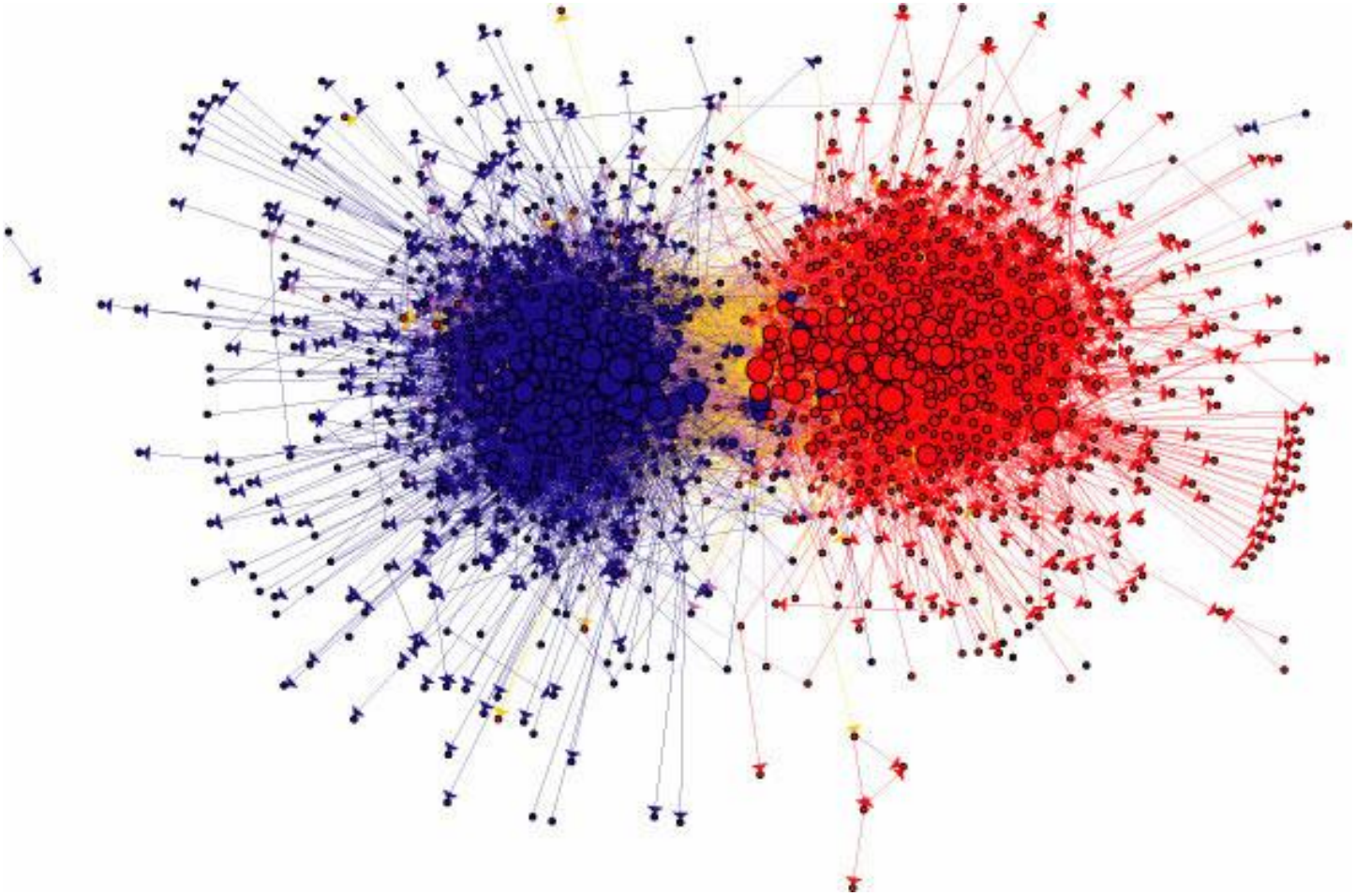
4-degrees of separation [Backstrom-Boldi-Rosa-Ugander-Vigna, 2011]

Biological Networks

- Genetic disorders and gene associations
- Drugs and their protein targets
- …



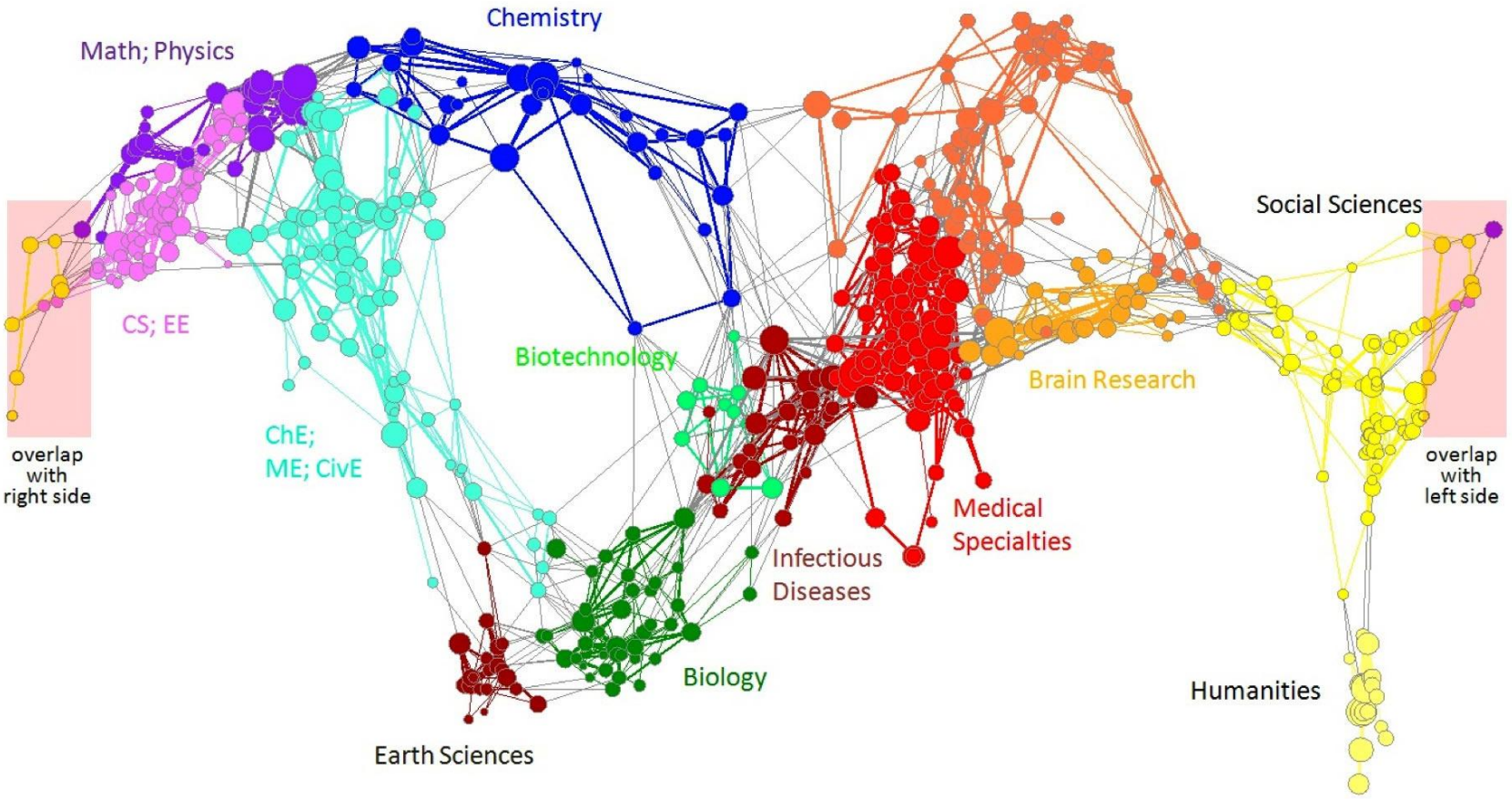
Media Networks



Connections between political blogs

Polarization of the network [Adamic-Glance, 2005]

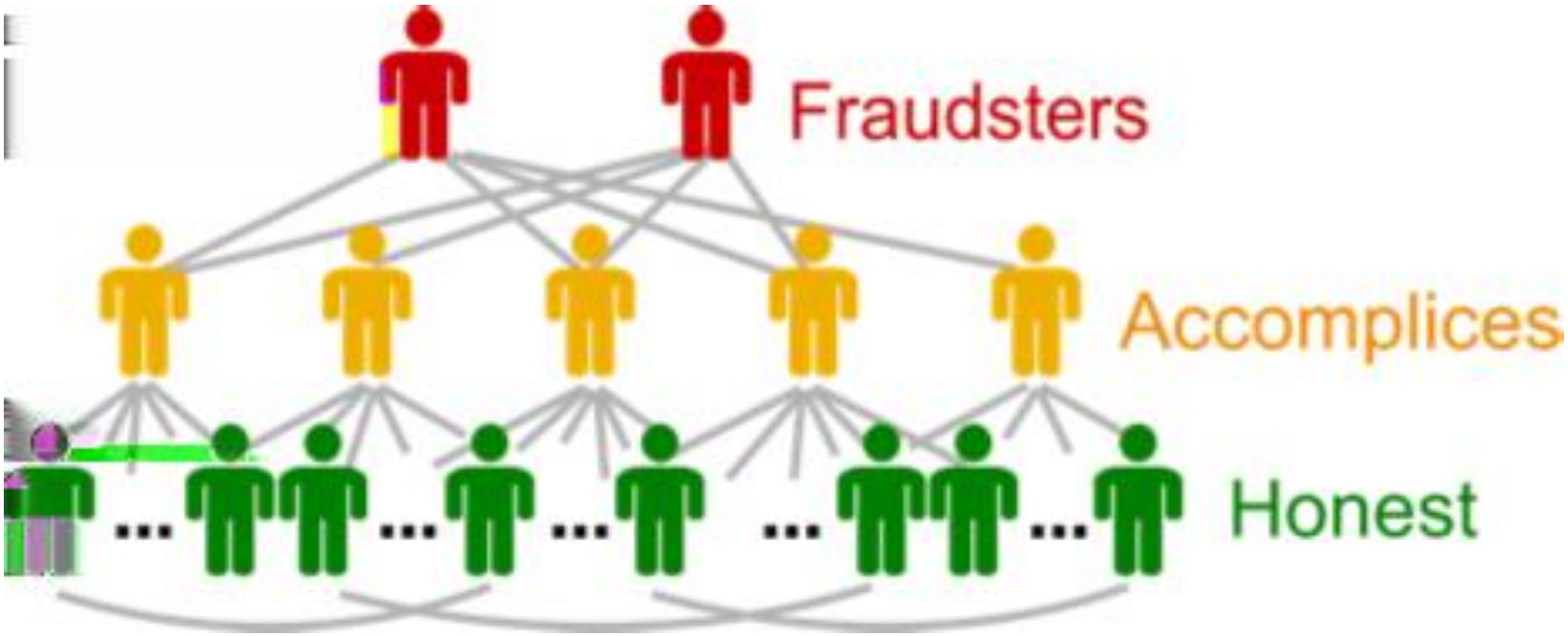
Graph Data: Information Nets



Citation networks and Maps of science

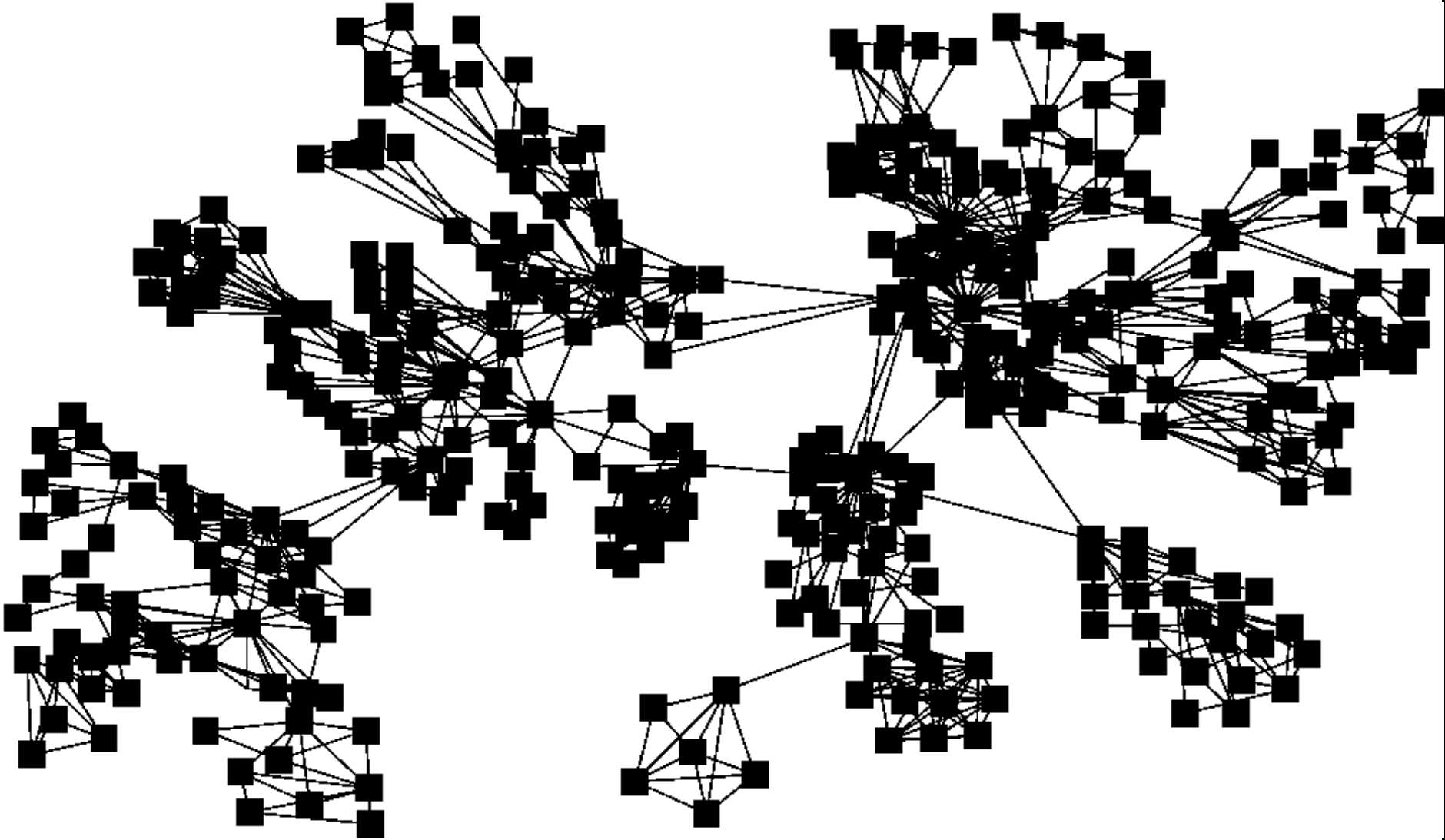
[Börner et al., 2012]

eBay Fraud Detection

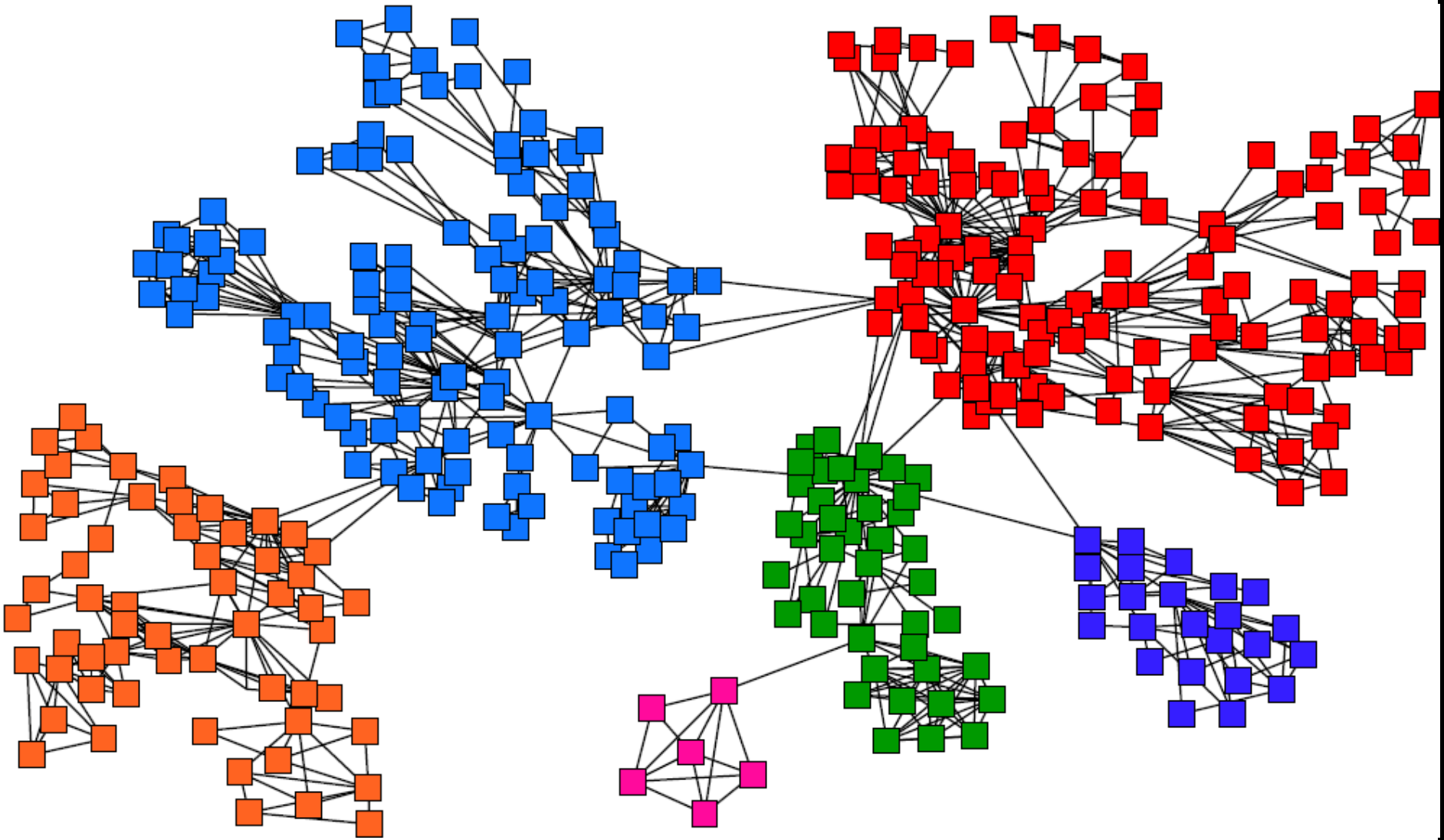


Communities

Goal: Find Densely Linked Clusters

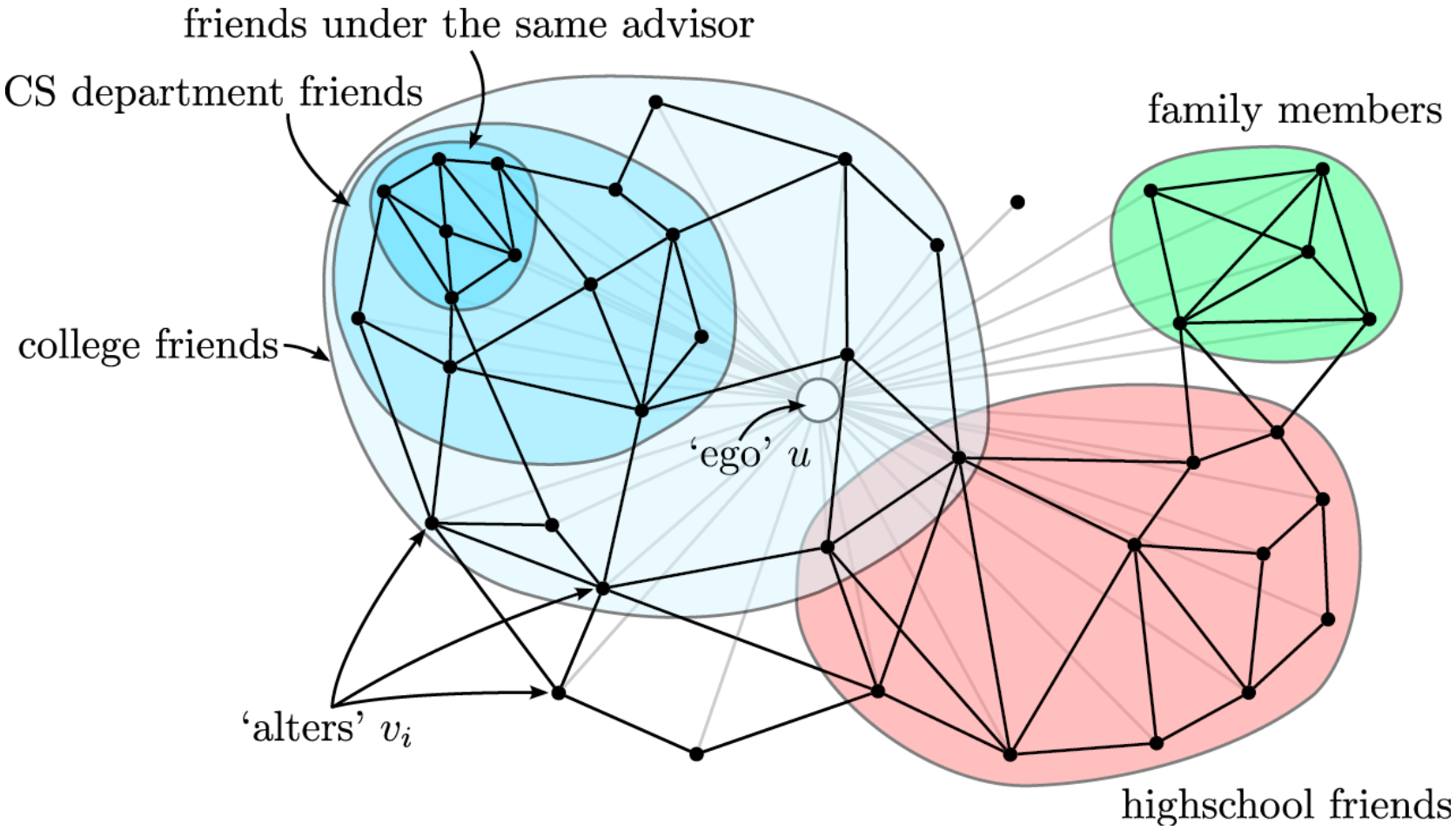


Goal: Find Densely Linked Clusters



Twitter & Facebook

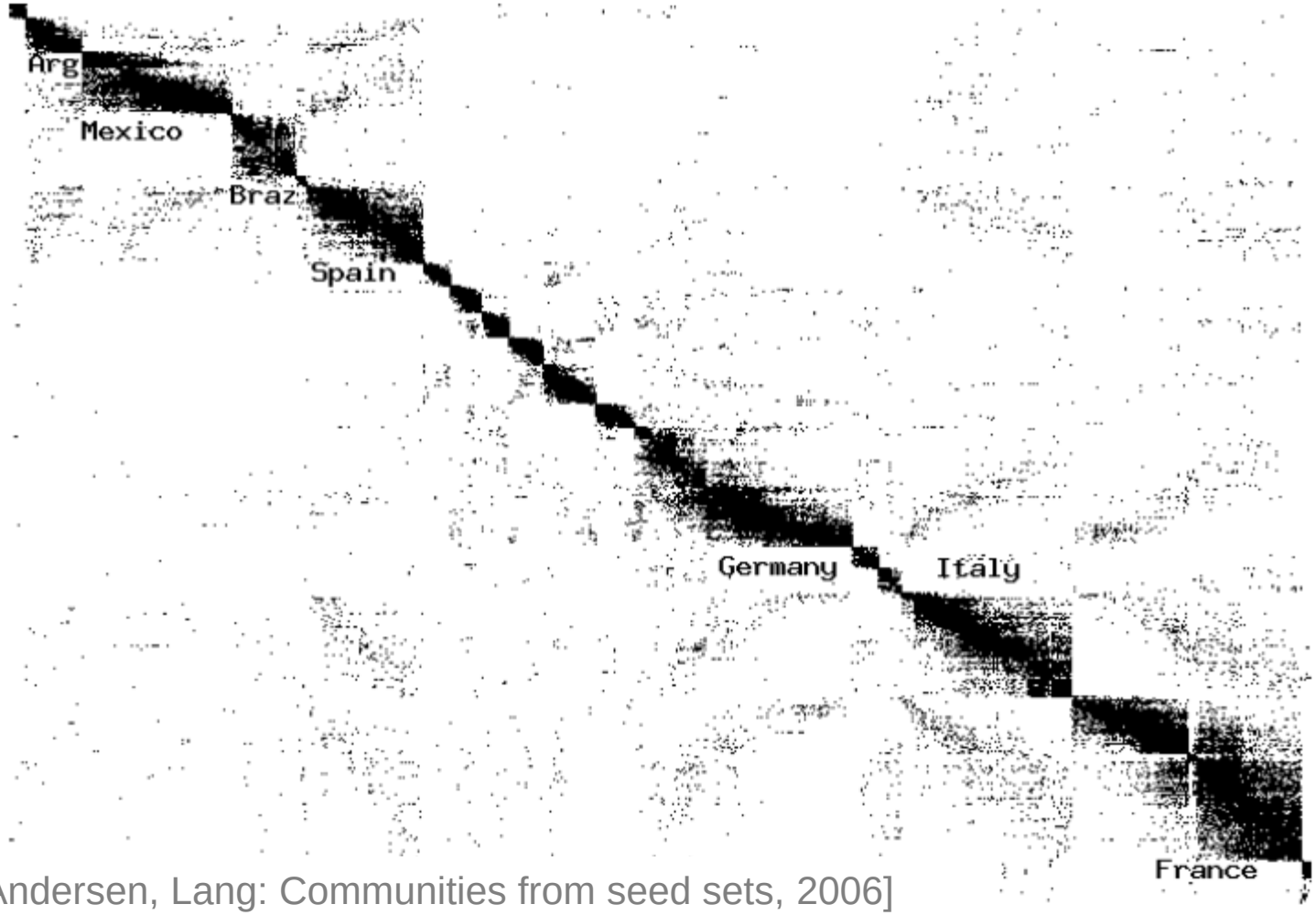
- Discovering social circles, circles of trust:



[McAuley, Leskovec: Discovering social circles in ego networks, 2012]

Movies and Actors

- Clusters in Movies-to-Actors graph:



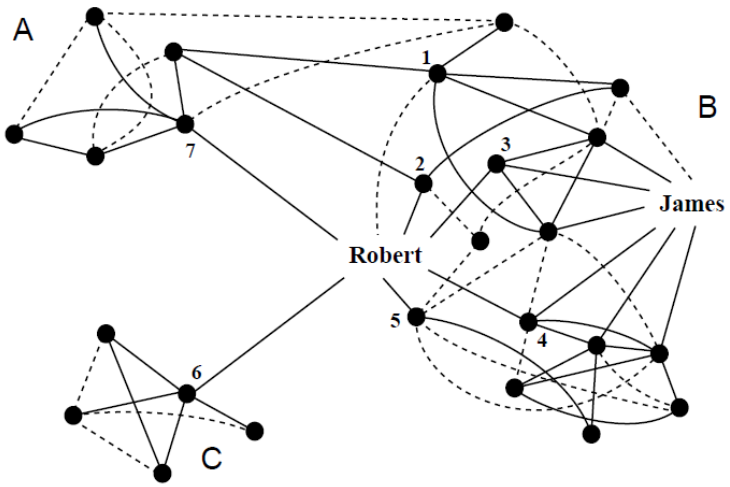
[Andersen, Lang: Communities from seed sets, 2006]

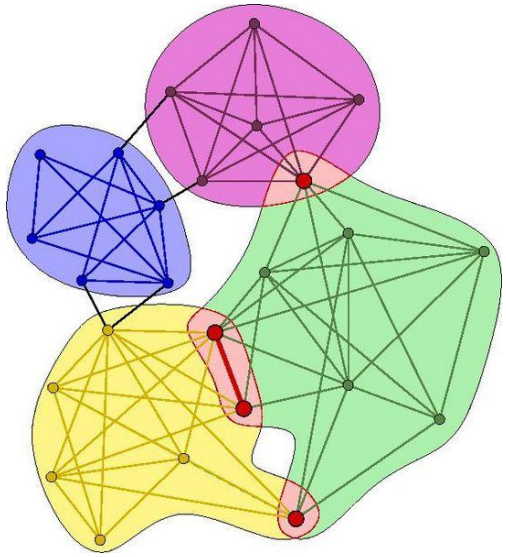
Community Detection

(undirected, unweighted networks)

Problem Definition

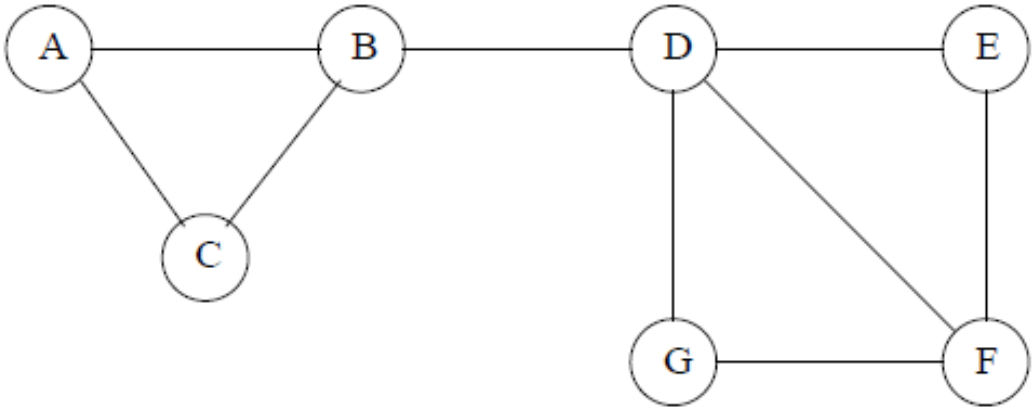
- Given an undirected, unweighted graph
- Find communities:
  - Subgraphs that are dense
  - Not many edges between communities





Why Different than Clustering?

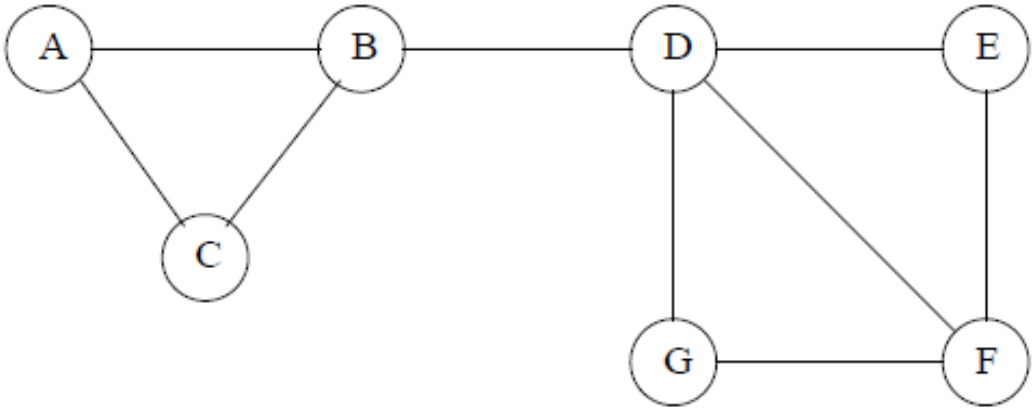
- Distance = graph distance



- Agglomerative clustering
  - –Edges are all same distance
  - –Pick any edge first

Why Different than Clustering?

- Distance = graph distance



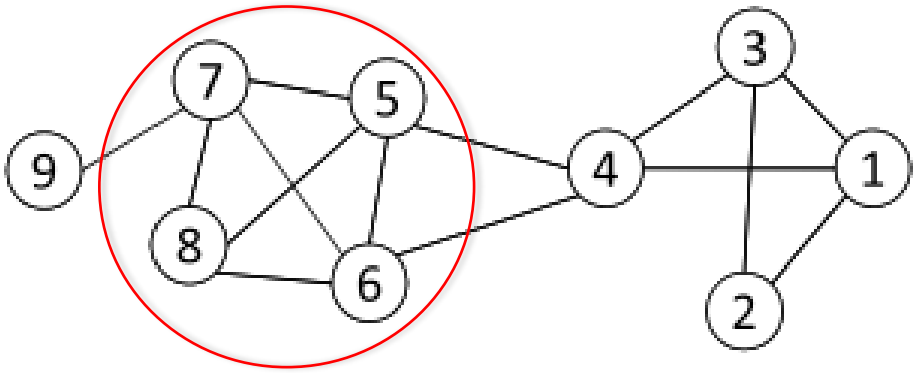
- Point assignment
  - –Starting nodes: B and F
  - –Where does D go?

How Would You Find Communities in a Graph?

- Subgraphs that are dense
- Not many edges between communities

Method 1: Cliques!

- Clique: a maximum complete subgraph in which all nodes are adjacent to each other



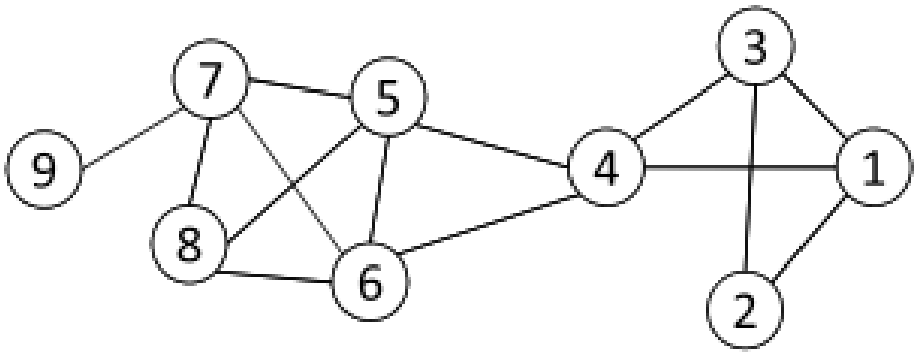
Nodes 5, 6, 7 and 8 form a clique

- NP-hard to find the maximum clique in a network
- Straightforward implementation to find cliques is very expensive in time complexity

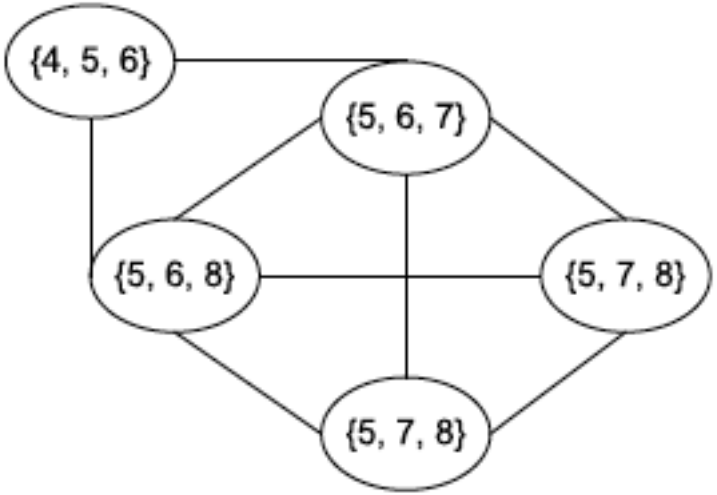
Clique Percolation

- Clique is a very strict definition, unstable
- Normally use cliques as a core or a seed to find larger communities
  - **Input**: A parameter k, and a network
  - **Procedure**
    - Find out all cliques of size k in a given network
    - Construct a clique graph. Two cliques are adjacent if they share k-1 nodes
    - Each connected components in the clique graph form a community
- Clique percolation finds overlapping communities

Clique Percolation Example



Cliques of size 3:
{1, 2, 3}, {1, 3, 4}, {4, 5, 6}, {5, 6, 7}, {5, 6, 8}, {5, 7, 8},
{6, 7, 8}



Communities:
{1, 2, 3, 4}
{4, 5, 6, 7, 8}

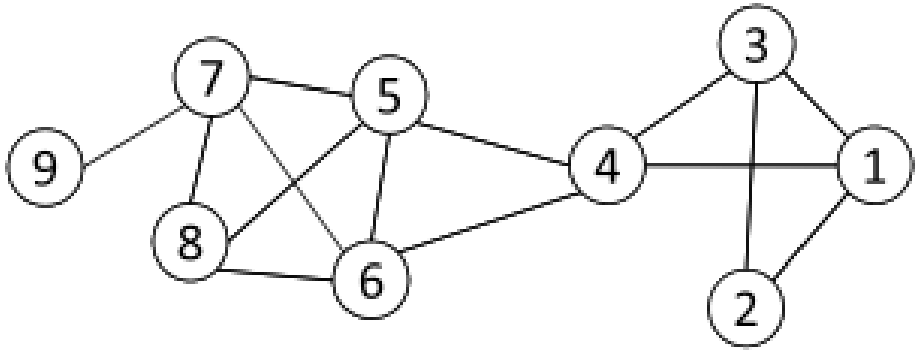
Problem with Clique Percolation

- Still expensive
- Assumes overlaps are not dense

Method 2: Clustering based on Vertex Similarity

- Apply k-means or similarity-based clustering to nodes
- Vertex similarity is defined in terms of the similarity of their neighborhood
- Structural equivalence: two nodes are structurally equivalent iff they are connecting to the same set of actors

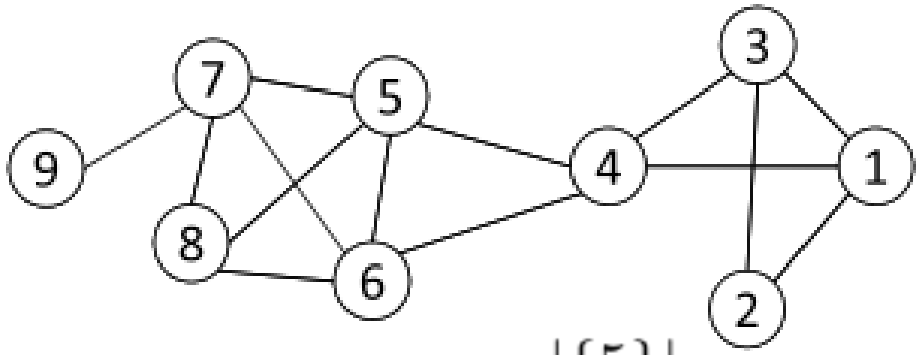
Nodes 1 and 3 are structurally equivalent;
So are nodes 5 and 6.



- Structural equivalence is too strict for practical use.

Vertex Similarity

- Jaccard Similarity $Jaccard(v_i, v_j) = \frac{|N_i \cap N_j|}{|N_i \cup N_j|}$
- Cosine similarity $Cosine(v_i, v_j) = \frac{|N_i \cap N_j|}{\sqrt{|N_i| \cdot |N_j|}}$



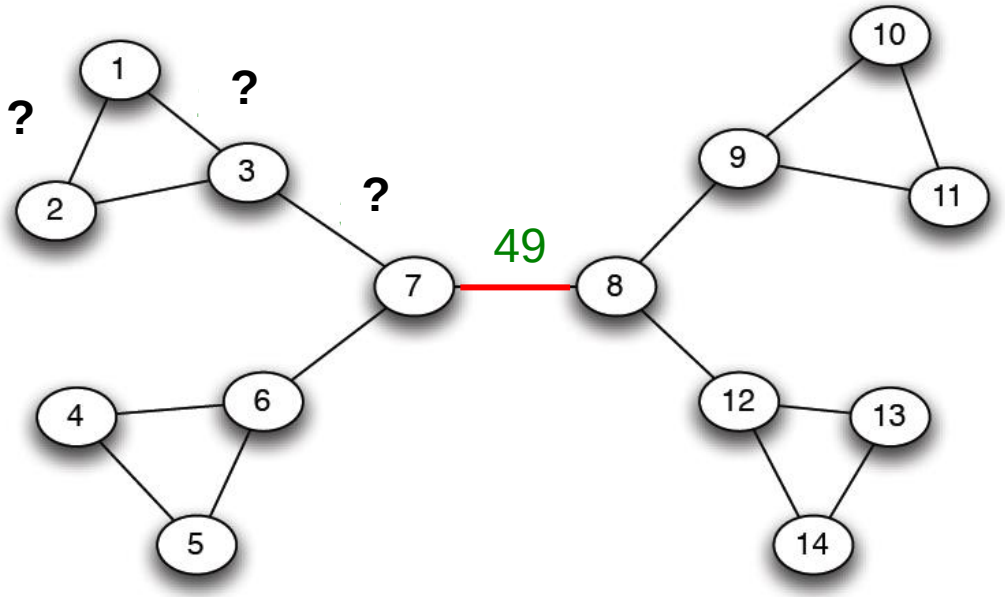
$$Jaccard(4,6) = \frac{|\{5\}|}{|\{1,3,4,5,6,7,8\}|} = \frac{1}{7}$$

$$cosine(4,6) = \frac{1}{\sqrt{4 \cdot 4}} = \frac{1}{4}$$

Method 3: Girvan-Newman

- Divisive hierarchical clustering based on the notion of edge betweenness:
  - *Number* of shortest paths passing through the edge
  - *Fraction* of shortest paths passing through the edge
    - Why interesting?

GIRVAN-NEWMAN: EXAMPLE

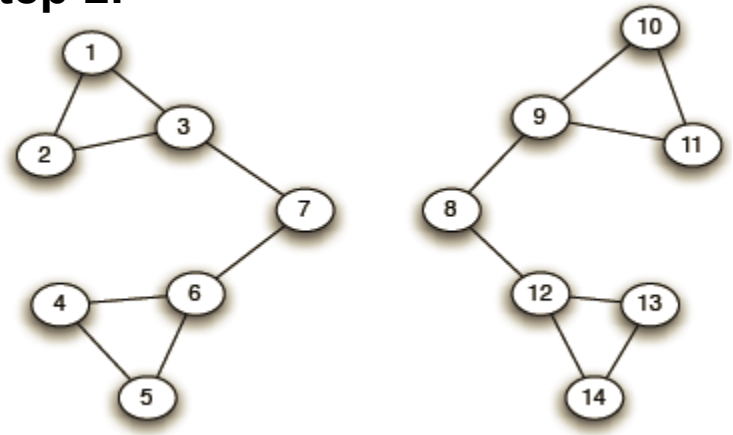


Method 3: Girvan-Newman

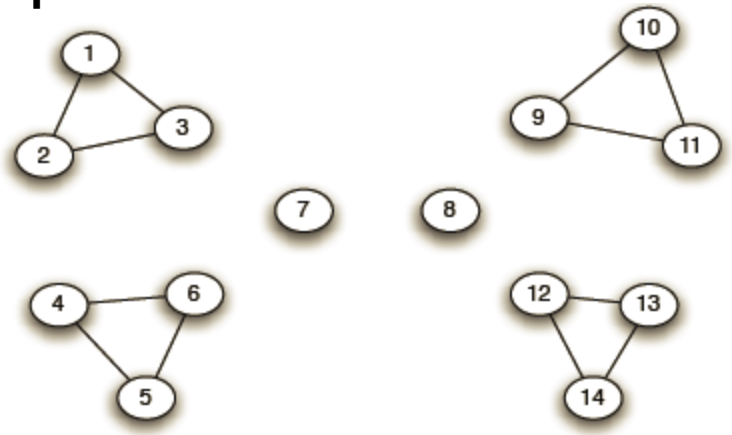
- Divisive hierarchical clustering based on the notion of edge betweenness:
  - *Number* of shortest paths passing through the edge
  - *Fraction* of shortest paths passing through the edge
- Girvan-Newman Algorithm:
  - **Repeat until no edges are left:**
    - Calculate betweenness of edges
    - Remove edges with highest betweenness
  - Connected components are communities
  - Gives a hierarchical decomposition of the network

GIRVAN-NEWMAN: EXAMPLE

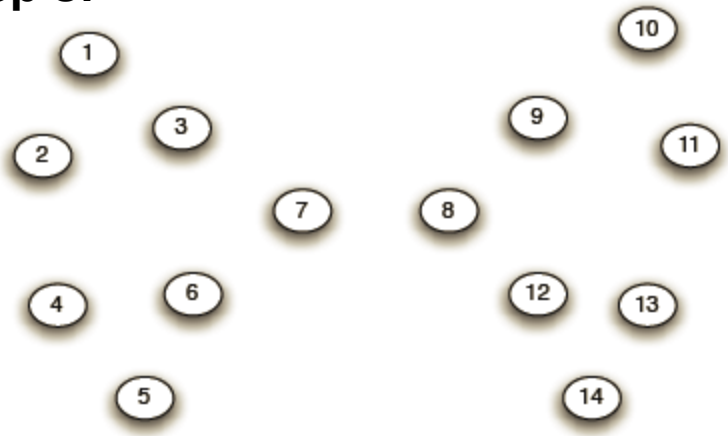
Step 1:



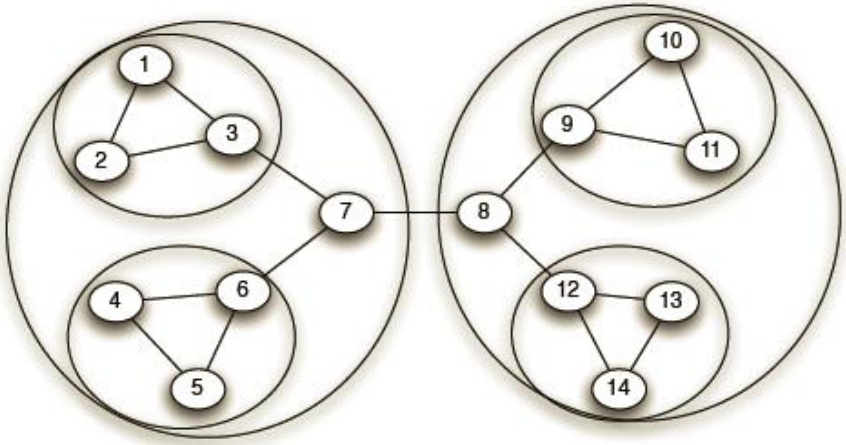
Step 2:



Step 3:

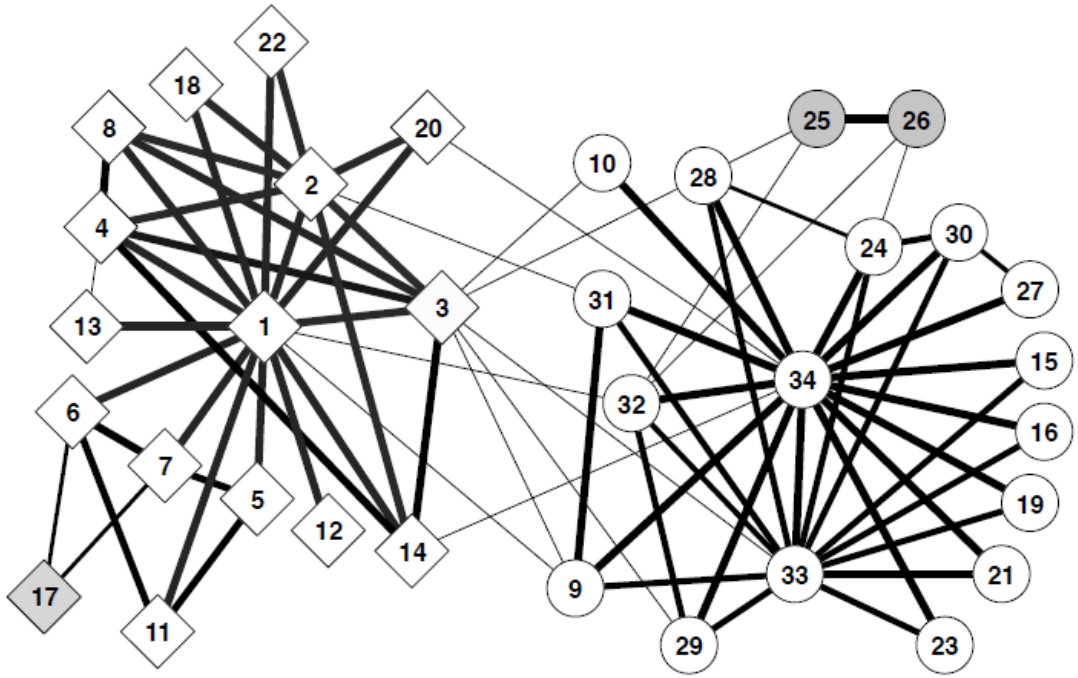


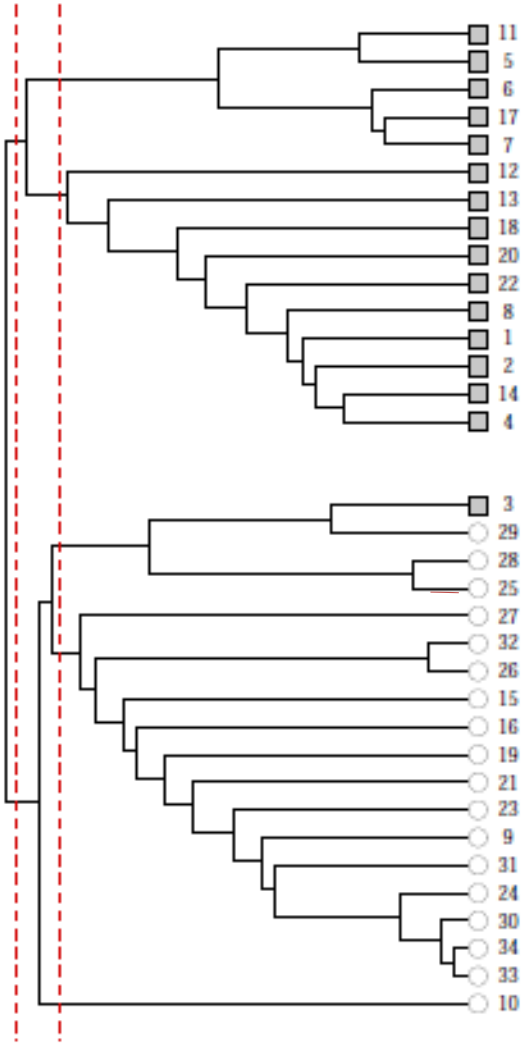
Hierarchical network decomposition:



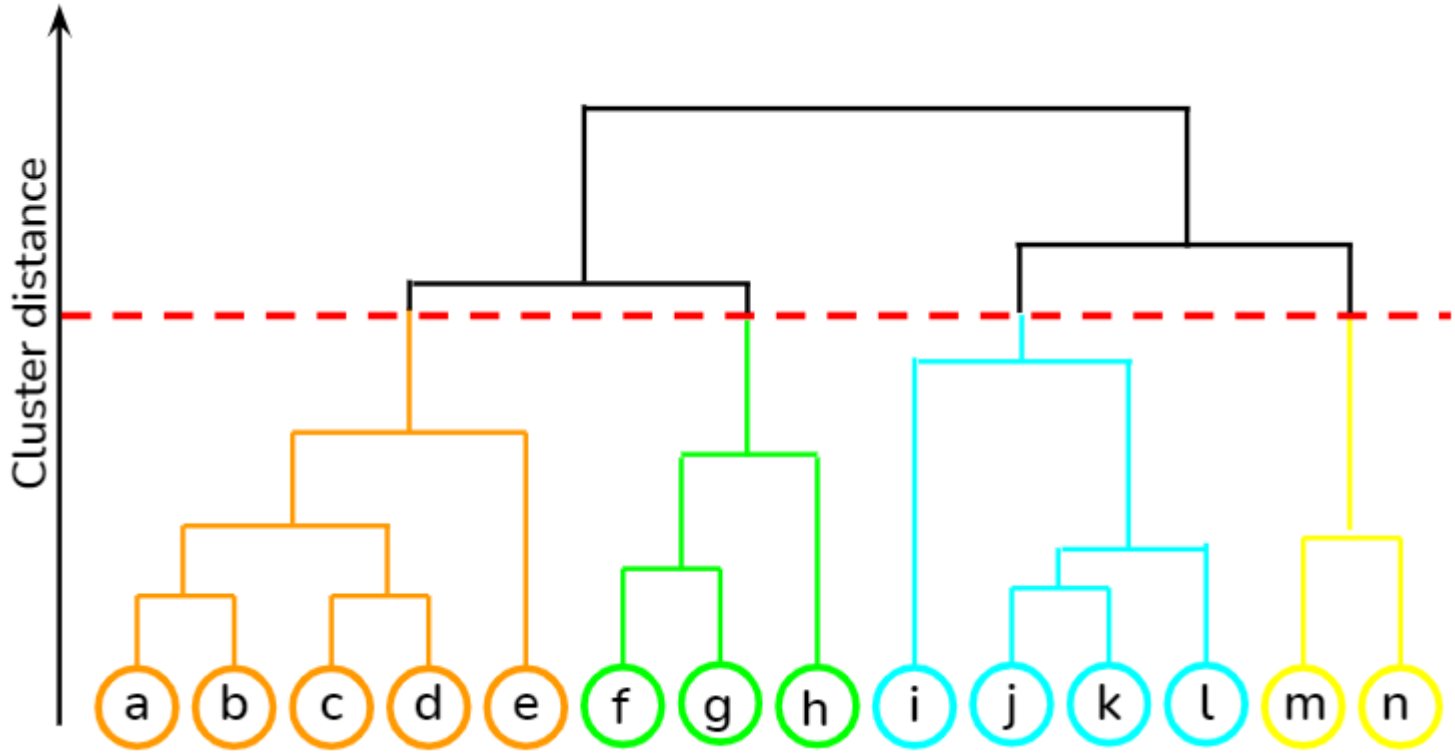
Girvan-Newman: Results

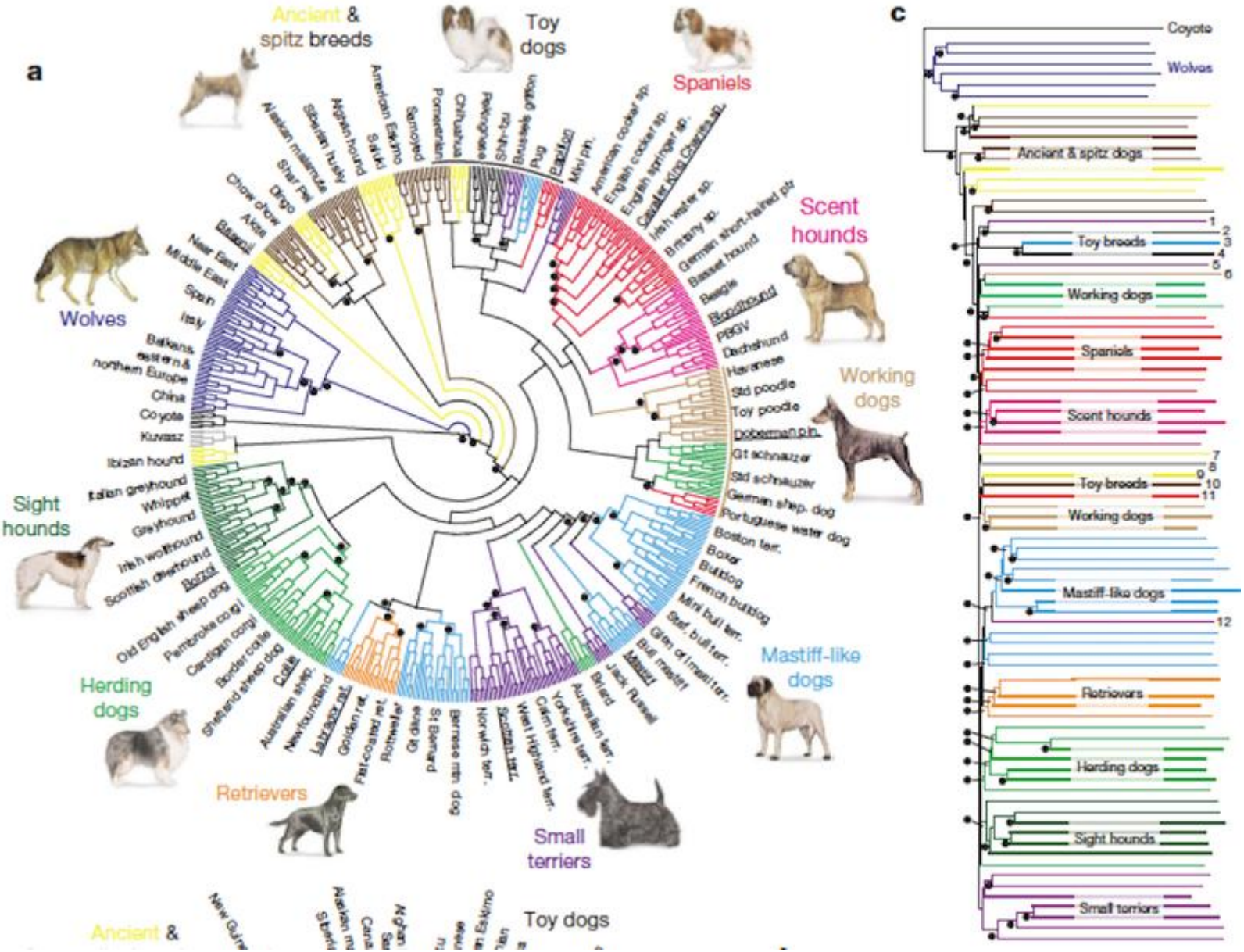
- Zachary's Karate club:
  Hierarchical decomposition





Dendrogram




a
Ancient & spitz breeds
Toy dogs
Spaniels
Scent hounds
Working dogs
Mastiff-like dogs
Retrievers
Small terriers
Herding dogs
Sight hounds
Wolves
Chihuahua
Pomeranian
Samoyed
American Eskimo
Saluki
Alaskan malamute
Siberian husky
Norwegian elkhound
Shar Pei
Dingo
Chow chow
Akita
Basenji
Near East
Middle East
Spain
Italy
Balkans, eastern & northern Europe
China
Coyote
Kuvasz
Ibizan hound
Italian greyhound
Whippet
Greyhound
Irish wolfhound
Scottish deerhound
Borzoi
Old English sheep dog
Pembroke corgi
Cardigan corgi
Border collie
Shetland sheep dog
Collie
Australian shep.
Newfoundland
Labrador ret.
Golden ret.
Flat-coated ret.
Rottweiler
Old shep.
St Bernard
Bernese mtn dog
Norwich terr.
Scottish terr.
West Highland terr.
Cairn terr.
Yorkshire terr.
Australian terr.
Bedlington terr.
Jack Russell
Miniat
Bull mastiff
Glen of Imaal terr.
Staf. bull terr.
Mini bull terr.
French bulldog
Bulldog
Boxer
Boston terr.
Portuguese water dog
German shep. dog
Std schnauzer
Gt schnauzer
Doberman pin.
Toy poodle
Std poodle
Havanese
Dachshund
PBGV
Bloodhound
Beagle
Basset hound
German short-haired pt
Brittany sp.
Irish water sp.
Cavalier King Charles sp.
English springer sp.
English cocker sp.
American cocker sp.
Mini pin.
Pug
Papillon
Brussels griffon
Shih-tzu
Pekingnese
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Toy dogs
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Coyote
Wolves
Ancient & spitz dogs
Toy breeds
Working dogs
Spaniels
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Mastiff-like dogs
Retrievers
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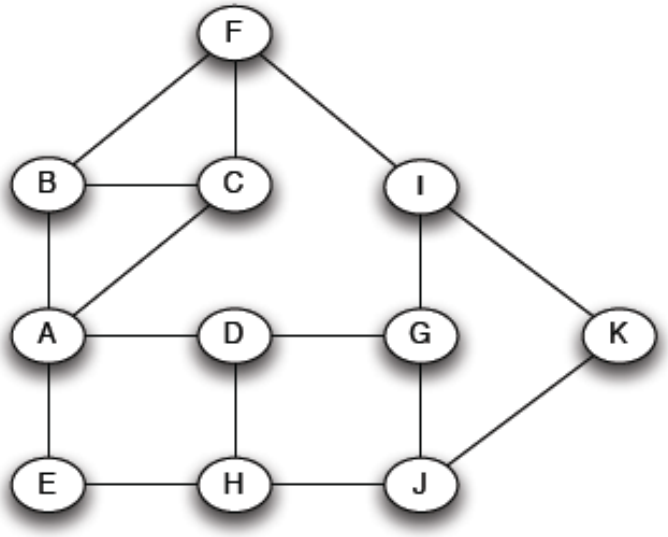
Two Questions Left

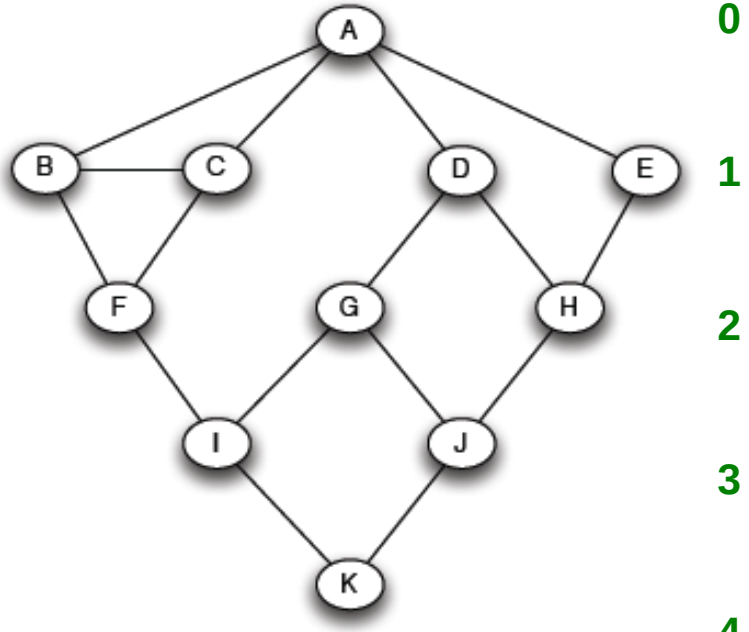
- How to compute betweenness?
- How to select the number of clusters?

How to Compute Betweenness?

- Want to compute betweenness of paths starting at node A

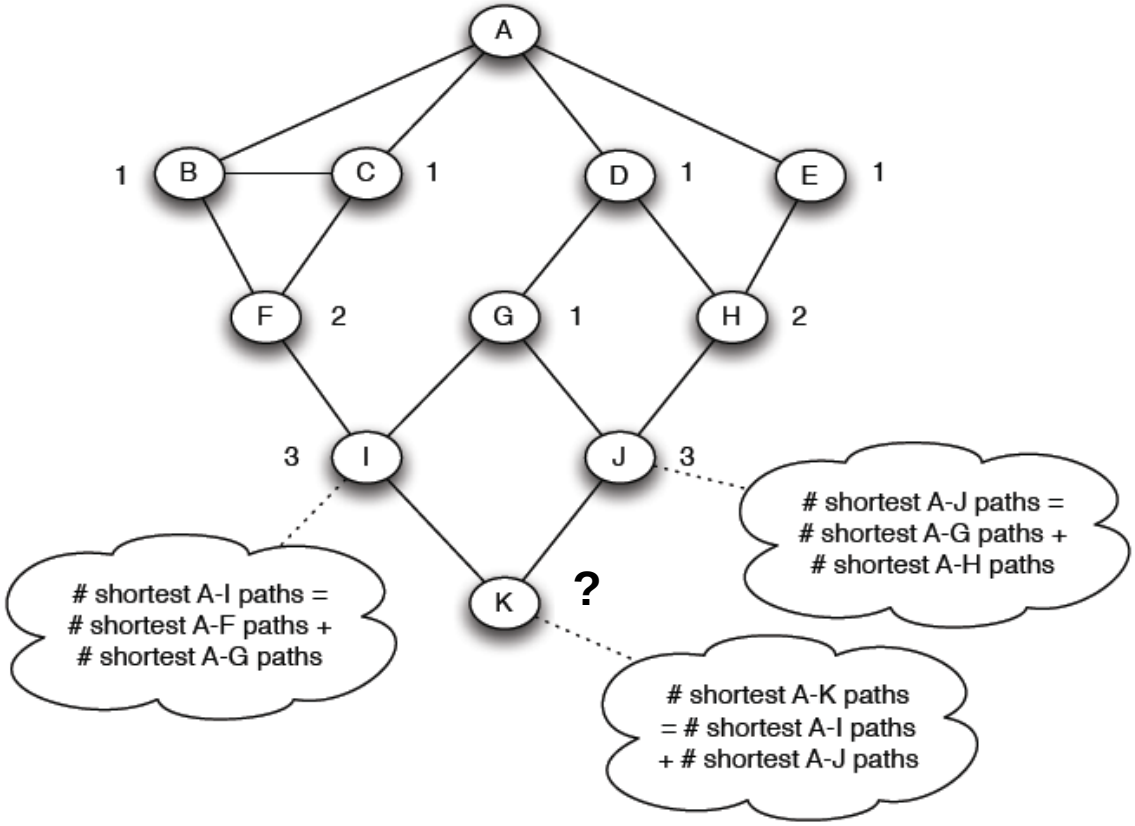


BFS starting from A:



How to Compute Betweenness?

- Count the number of shortest paths from A to all other nodes of the network:



How to Compute Betweenness?

- Compute fractional betweenness by working up the tree: If there are multiple paths count them fractionally

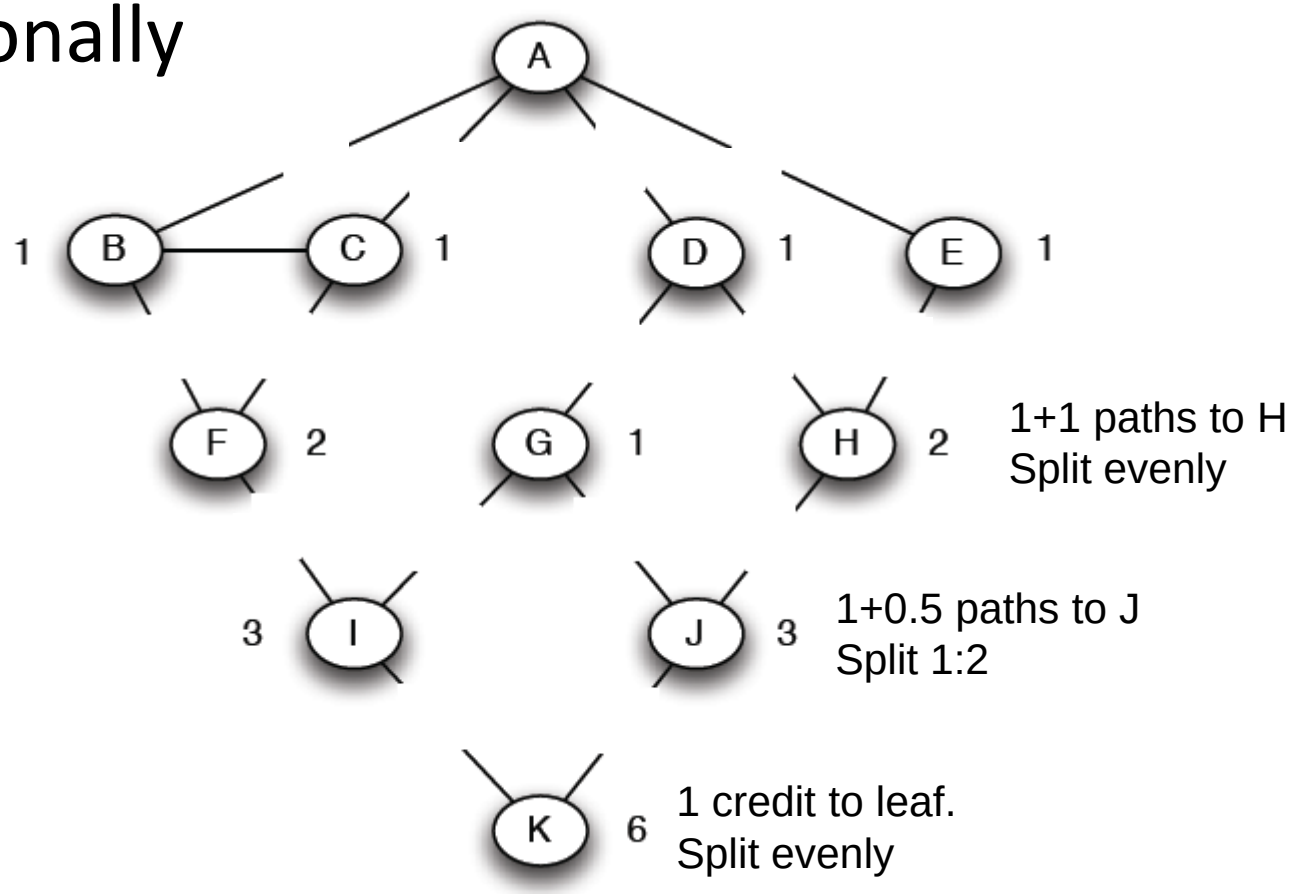
The algorithm:

•Add edge **flows**:

-- node flow = 1+∑child edges

-- split the flow up based on the parent value

• Repeat the BFS procedure for each starting node U



How to Compute Betweenness?

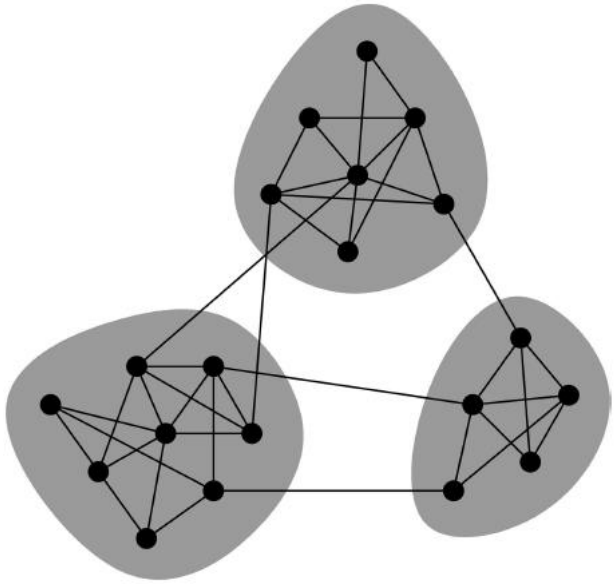
- Sum credits for each edge
- Divide by 2
  - Each shortest path discovered twice (why?)

Two Questions Left

- How to compute betweenness?
- How to select the number of clusters?

Network Communities

- **Communities:** sets of tightly connected nodes
- Define: **Modularity Q**
  - A measure of how well a network is partitioned into communities
  - Given a partitioning of the network into groups $s \in S$:



$$Q \propto \sum_{s \in S} [\ (\#\ \text{edges within group } s) - (\text{expected } \#\ \text{edges within group } s)\ ]$$

Need a null model!

Null Model: Configuration Model

- **Given real G on n nodes and m edges, construct rewired network G'**

  –Same degree distribution but random connections

  - Break each edge into two "stubs"
  - Consider G' as a multigraph

Null Model: Configuration Model

- **The expected number of edges between nodes i and j** of degrees k_i and k_j equals to: $k_i \cdot \frac{k_j}{2m}$

Note:

$$\#stubs = \sum_{u \in N} k_u = 2m$$

Modularity

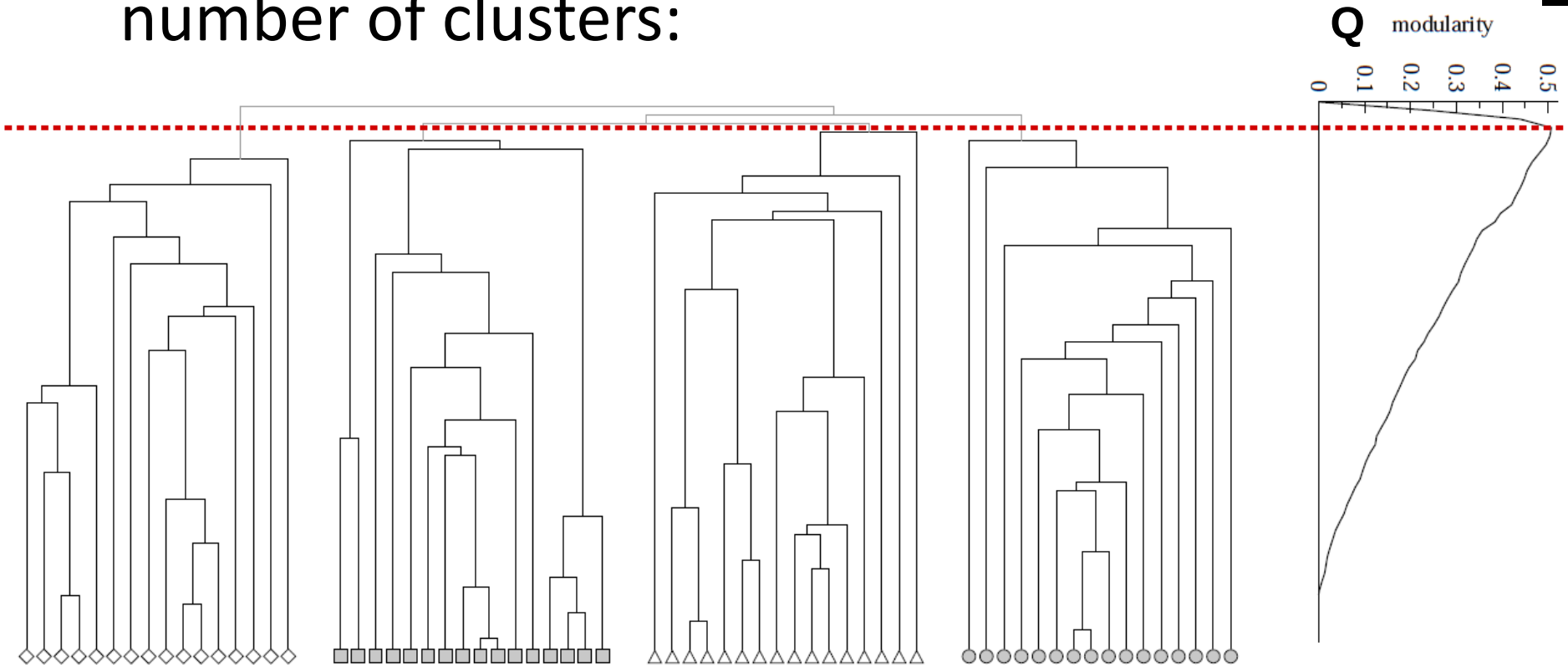
- **Modularity of partitioning S of graph G:**
  - **$Q \propto \sum_{s \in S}$ [(# edges within group s) – (expected # edges within group s)]**
  - $Q(G,S) = \frac{1}{2m} \sum_{s \in S} \sum_{i \in s} \sum_{j \in s} \left(A_{ij} - \frac{k_i k_j}{2m} \right)$

A_{ij} = 1 if i→j, 0 else

- **Modularity values take range [−1,1]**
  - It is positive if the number of edges within groups exceeds the expected number
  - **0.3-0.7<Q** means significant community structure

Modularity: Number of clusters

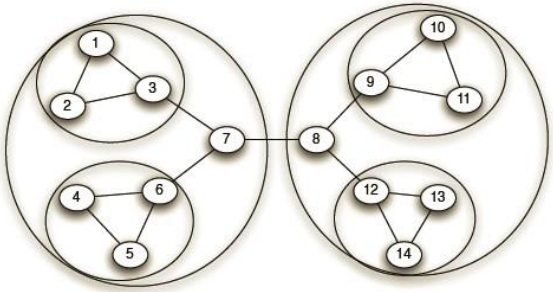
- Modularity is useful for selecting the number of clusters:



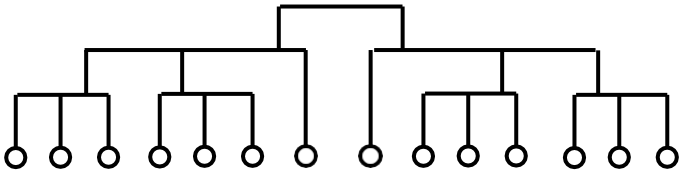
Louvain Algorithm

- **Greedy algorithm** for community detection
  - $O(n \log n)$ run time
- Supports weighted graphs
- Provides hierarchical communities
- Widely utilized to **study large networks** because:
  - Fast
  - Rapid convergence
  - High modularity output (i.e., “better communities”)

Network and communities:



Dendrogram:

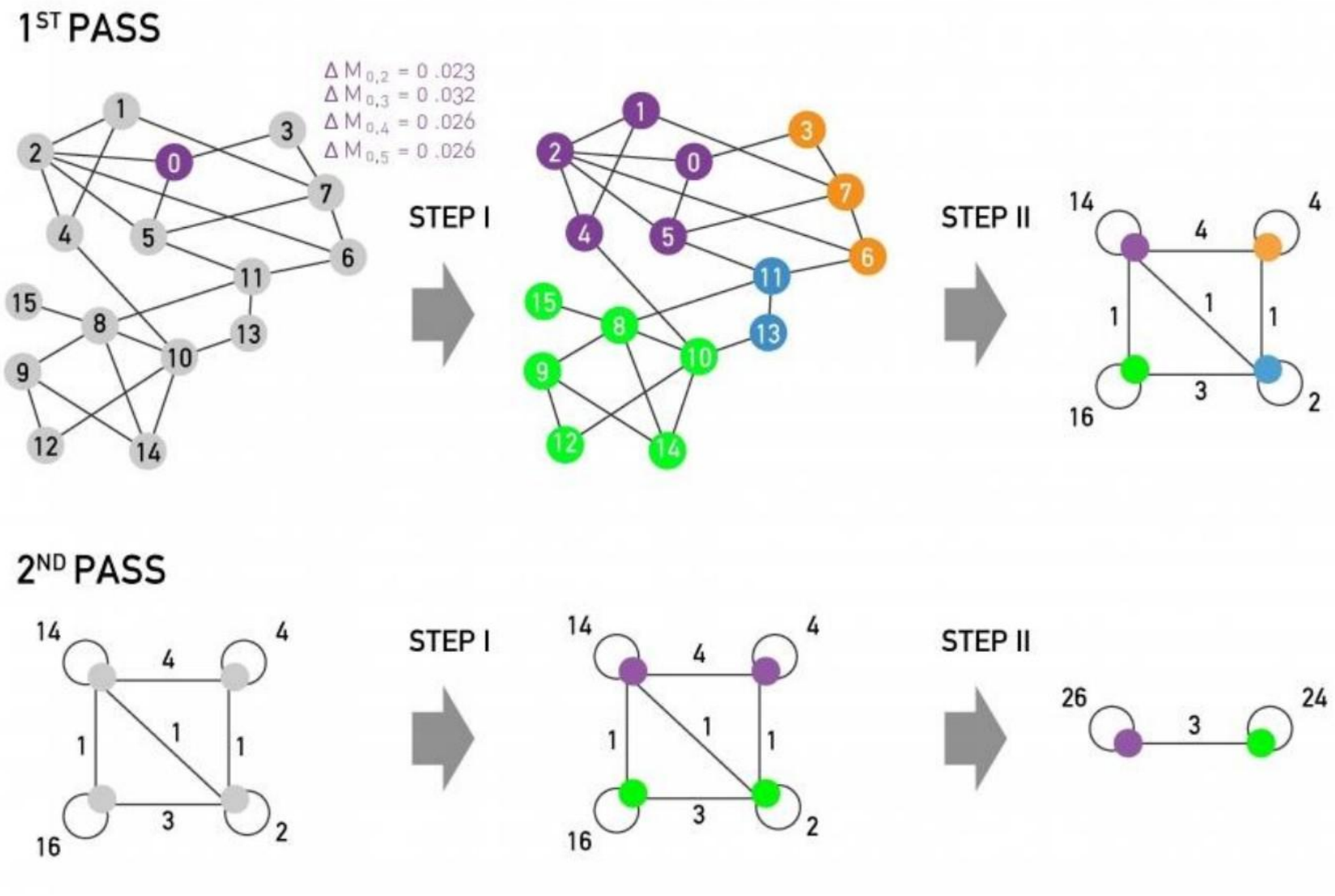


Louvain Algorithm: At High Level

- Louvain algorithm **greedily maximizes** modularity
- **Each pass is made of 2 phases:**
  - **Phase 1:** Modularity is optimized by allowing only local changes to node-communities memberships
  - **Phase 2:** The identified communities are aggregated into super-nodes to build a new network
  - **Goto Phase 1**

The passes are repeated **iteratively** until no increase of modularity is possible.

Louvain Algorithm



Louvain: 1st phase (Partitioning)

- Put each node in a graph into a distinct community (one node per community)
- For each node i, the algorithm performs two calculations:
  - Compute the modularity delta (ΔQ) when putting node i into the community of some neighbor j
  - Move i to a community of node j that yields the largest gain in ΔQ
- **Phase 1 runs until no movement yields a gain**

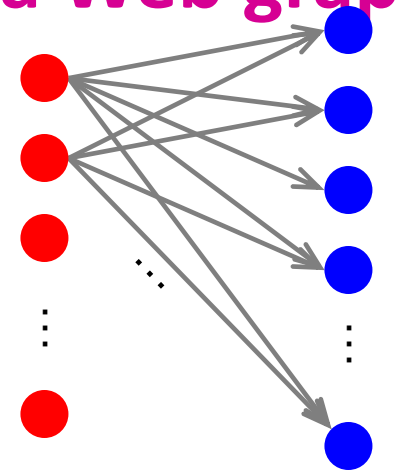
Louvain: 2nd phase (Restructuring)

- The communities obtained in the first phase are contracted into **super-nodes**, and the network is created accordingly:
  - Super-nodes are connected if there is at least one edge between the nodes of the corresponding communities
  - The weight of the edge between the two super-nodes is the sum of the weights from all edges between their corresponding communities
- **Phase 1 is then run on the super-node network**

Trawling

Trawling

- **Searching for small communities in the Web graph**
- **What is the signature of a community / discussion in a Web graph?**



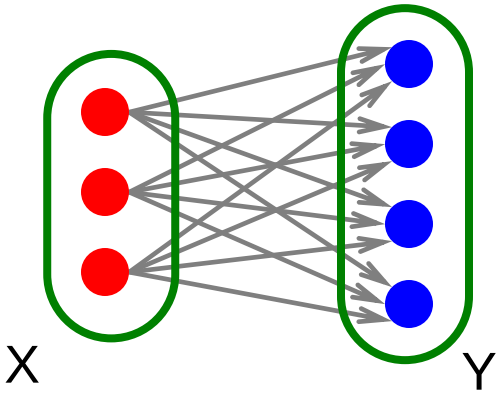
Dense 2-layer graph

Use this to define "topics":
What the same people on the left talk about on the right

Intuition: Many people all talking about the same things

Searching for Small Communities

- **A more well-defined problem:**
  Enumerate complete bipartite subgraphs $K_{s,t}$
  - Where $K_{s,t}$: s nodes on the "left" where each links to the same t other nodes on the "right"



$K_{3,4}$

|X| = s = 3
|Y| = t = 4

Fully connected

COMMUNITY EVALUATION

Evaluating Community Detection (1)

- For groups with clear definitions
  - E.g., Cliques, k-cliques, k-clubs, quasi-cliques
  - Verify whether extracted communities satisfy the definition
- For networks with ground truth information
  - Accuracy of pairwise community memberships

Accuracy of Pairwise Community Memberships

- Consider all the possible pairs of nodes and check whether they reside in the same community
- An error occurs *if*
  - Two nodes belonging to the same community are assigned to different communities
  - Two nodes belonging to different communities are assigned to the same community
- Construct a contingency table or confusion matrix

		Ground Truth	
		$C(v_i) = C(v_j)$	$C(v_i) \neq C(v_j)$
Clustering Result	$C(v_i) = C(v_j)$	a	b
	$C(v_i) \neq C(v_j)$	c	d

$$accuracy = \frac{a+d}{a+b+c+d} = \frac{a+d}{n(n-1)/2}$$

Accuracy Example

Ground Truth

Algorithm Result

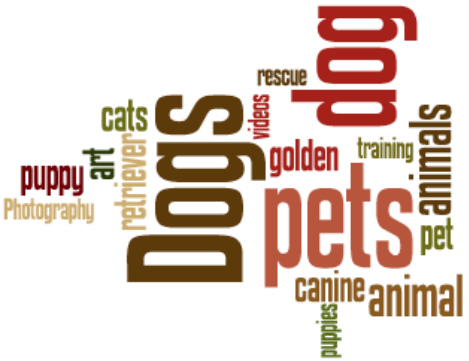
		Ground Truth	
		$C(v_i) = C(v_j)$	$C(v_i) != C(v_j)$
Algorithm Result	$C(v_i) = C(v_j)$		
	$C(v_i) != C(v_j)$		

Accuracy =

Evaluation using Semantics

- For networks with semantics
  - Networks come with semantic or attribute information of nodes or connections
  - Human subjects can verify whether the extracted communities are coherent
- Evaluation is qualitative
- It is also intuitive and helps understand a community

An *animal* community



A *health* community

Evaluation without Ground Truth

- For networks without ground truth or semantic information
- This is the most common situation
- Quantitative evaluation functions
  - Modularity
  - Link prediction (the predicted network is compared with the true network)